

ATM Network Security Protocol

Module 29

Duration: 25:45

Video: <https://www.youtube.com/watch?v=JdRurgWuHvQ>

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0:01 1 second [Music]

0:17 Hello friends I welcome you all to the video lecture series of cyber security and information security in this lecture

0:25 we will study about ATM network security protocol so what is an ATM asynchronous transfer

0:34 mode ATM is a switching technique used by telecommunication networks that uses asynchronous time division multiplexing

0:42 to encode data into small fixed sized cells this is different from Ethernet or Internet which use variable packet sizes

0:51 for data or frames ATM is the core protocol used over the synchronous Optical Network Sonet backbone of the

1:00 ISDN asynchronous transfer mode was designed with sales in mind this is because voice data is converted to

1:07 packets and is forced to share a network with burst data passing through the same medium so no matter how small the voice

1:15 packets are they always encounter full size data packets and could experience maximum queing

1:23 delays this is why all data packets should be of the same size the fixed Sal structure of of ATM means it can be

1:31 easily switched by Hardware without the delays introduced by rooted frames and software switching this is why some

1:38 people believe that ATM is the key to the internet bandwidth problem ATM creates fixed Roots between two points

1:47 before data transfer begins which differs from tcpip where data is divided into packets Each of which takes a

1:54 different route to get to its destination this makes it easier to build data usage however an ATM network

2:03 is less adaptable to a sudden Network traffic search the ATM provides data link layer services that run on the OSI

2:11 layer one physical links it functions much like small packet switched and circuit switch networks which makes it

2:19 ideal for real time low latency data such as viip and video as well as for high

2:26 throughput data traffic like file transfers a virtual circuit or connection must be established before

2:33 the two end points can actually exchange data ATM Services generally have four different bit rate choices available bit

2:42 rate provides a guaranteed minimum capacity but data can be bursted to higher capacities when Network traffic

2:49 is minimal constant bit rate specifies a fixed bit rate so that data is sent in a

2:56 steady stream this is analogous to a leased line unspecified bit rate doesn't guarantee

3:03 any throughput level and is used for applications such as file transfers that can tolerate delays variable bit rate

3:12 vbr provides a specified throughput but data is not sent evenly this makes it a even popular choice for voice and video

3:21 conferencing the structure of an ATM cell an ATM cell consists of a 5 byte

3:27 header and a 48 by payload ATM defines two different cell formats as shown in

3:34 the figure uni user network interface and nni that is network network

3:41 interface most ATM links use uni cell format GFC that is generic flow control

3:50 four bits and default is 40 bits VPI that is virtual path identifier

3:58 8 Bits uni or 12 Bits n ni I VCI virtual

4:03 Channel identifier 16 bits PT P type that is three bits PT bit three Network

4:13 management cell if zero user data cell and the following apply PT bit 2

4:21 explicit forward conen indication efc1 1 is equal to network congestion

4:28 experienced PT bit 1 On's bit ATM user to user AA bit used by aal5 to indicate

4:38 packet boundaries CLP that is cell loss priority 1

4:46 bit HC that is header error control which is

4:51 8bit CRC polinomial is equal to $x^8 + x^2 + 1$

5:00 ATM uses the PT field to designate various special kinds of sales for operations Administration and management

5:08 that is om purposes and to delineate packet boundaries in some ATM adaptation layers A

5:16 A if the most significant bit of the PT field is zero this is a user data cell and the other two bits are used to

5:25 indicate Network congestion and as a general purpose header bit available for ATM adaptation

5:31 layers if the MSB bit of the PT bit is one this is a management cell and the other two bits indicate the type that is

5:40 Network management segment Network management end to end resource management and reserved for future

5:48 use several ATM link protocols use the HC field to drive a CRC based framing

5:55 algorithm which allows locating the ATM cells with no overhead beyond what is otherwise needed for header protection

6:04 the 8bit CRC is used to correct single bit header errors and detect multi-bit header errors when multi-bit header

6:12 errors are detected the current and subsequent sales are dropped until a cell with no header errors is

6:19 found a un cell reserves the GFC field for a local flow controls sub

6:26 multiplexing system between users this was intended to allow several terminals to share a single network connection in

6:34 the same way that two integrated Services digital Network ISDN phones can share a single basic rate ISDN

6:42 connection all four GFC bits must be Zero by default the nni cell format replicates

6:51 the uni format almost exactly except that the 4bit GFC field is

6:57 reallocated to the V field extending the VPI to 12 bits thus the single nni ATM

7:06 interconnection is capable of addressing almost 2^{16} VPS of up to almost 2^{16} VCS each in

7:16 practice some of the VP and VC numbers are reserved ATM network interfaces following are the ATM network interfaces

7:25 and are shown in the figure user to network interface public network interface private network

7:33 interface second Network to node interface which includes private nni public

7:40 nni data exchange interface dxi between packet routers and ATM digital service

7:48 units DSU the ATM adaption layer how to break

7:54 application messages to sales the ATM layer transmission switching

8:01 8 minutes, 1 second reception congestion control buffer management cell header generation removal at source or

8:09 destination reset connection identifiers for the next hope at switch sell address translation

8:17 sequential delivery the ATM Forum specifications data confidentiality the

8:24 function of confidentiality is to protect precious business data from unauthorized persons confidentiality

8:32 part of network security makes sure that the data is available only to the intendent and authorized

8:39 persons data integrity and authentication this goal means maintaining and assuring the accuracy

8:47 and consistency of data the function of Integrity is to make sure that the data is

reliable and is not changed by

- 8:55 unauthorized persons AAL frames are protected Ed by appending a cryptographic check sum to a
- 9:03 frames possibility is given to provide reordering protection by introducing a sequence number into AAL frames before
- 9:12 calculating the cryptographic check sum signaling protection any signaling messages may be authenticated and
- 9:20 integrity protected by introducing a digital signature into an siie especially if anme protocol is
- 9:28 employed protection is offered for setup and connects messages by calculating the signature over the siie fields specified
- 9:37 by the smme protocol it is still offered for any other signaling messages release
- 9:44 status restart and for setup and connect messages if the smme protocol is not used by introducing a signature
- 9:53 calculated over part of the siie availability this goal means maintaining and assuring
- 10:02 the accuracy and consistency of data the function of Integrity is to make sure that the data is reliable and is not
- 10:09 changed by unauthorized persons so ATM security system should provide the following functions verification of
- 10:18 identities security system should be able to establish and verify the claimed identity of any actor in an ATM
- 10:25 network controlled access and authorization the actors should not be able to gain access to information or resources if
- 10:34 they are not authorized to protection of confidentiality stored and communicated
- 10:41 data should be confidential protection of data Integrity the security system should guarantee the Integrity of the
- 10:49 stored and communicated data strong accountability an entity cannot deny the responsibility of its performed actions
- 10:57 as well as their effects activities logging the security system should support the capability to
- 11:05 retrieve information about security activities in the network elements with the possibility of tracing this
- 11:11 information to individuals or entities alarm reporting the security system should be able to generate alarm
- 11:20 notification about certain adjustable and selective security related events audit when violations of security happen
- 11:29 the system should be able to analyze the logged data relevant to security security recovery the security system

11:36 should be able to recover from successful or attempted breaches of security Security

11:43 Management the security system should be able to manage the security services derived from the above requirements they

11:51 are necessary to support the maintenance of Security Services if the security system cannot be recovered from attacks

11:58 and not providing Security Services anymore then the system won't be secure after these attacks on the other hand

12:05 Security Services and information about security have to be managed securely they are foundations of the security

12:13 system threads to ATM as other networks ATM networks will suffer a lot of threats typical ones are

12:23 eavesdropping spoofing service denial VC stealing and traffic analysis

12:30 Etc notice that VC stealing and traffic analysis happen only in ATM networks

12:38 eavesdropping eavesdropping refers to the threat that the attacker connects or taps into the transmission media and

12:45 gain unauthorized access to data it is one of the most common attacks to the network since most ATM networks are

12:52 connected with fiber optic cables some people might get the wrong impression that it is not so easy to tap a ATM

13:00 network spoofing spoofing attack means that an attacker tries to impersonate another

13:08 user to the third part therefore can get access to resources belonging to the victim to take advantage or just destroy

13:16 them spoofing might need special tools to manipulate the protocol data unit sometimes it might require the attacker

13:23 has special access permission C must be super user in Unix environment however since a network will be

13:30 connected to many interested networks via the Internet it's impossible to prevent a hacker from getting this excess

13:38 permission or even trace the people with this particular excess permission ATM is being implemented in public domain

13:45 therefore it is subjected to this kind of attack also service denial ATM is a

13:52 connection oriented technique a connection which is called virtual circuit VC in ATM is managed by a set of

14:01 14 minutes, 1 second signals VC is established by setup signals and can be disconnected by release or drop party signals if an

14:11 attacker sends release or drop party signal to any intermediate switch on the way of a VC then the VC will be

14:19 disconnected by sending these signals frequently the attacker can greatly disturb the communication between one

14:26 user to another therefore will disable the quality of service in ATM combining this technique with other tricks like

14:34 eavesdropping the attacker can even completely block one user from another stealing of VCS if two switches

14:44 in an ATM network compromise the attacker can even steal a VC from another user say vc1 and vc2 are two

14:53 virtual channels which will go through switch a and switch B wec one is owned

14:59 by user U1 and vc2 is owned by user U2 if a and b have compromised then a can

15:06 switch VC one's cell going from A to B through vc2 and B will switch back those

15:13 cells to vc1 since switches will forward cells based on the VCI that is virtual Channel identifier or VPI that is

15:22 virtual path identifier in the cell header A and B can just alter these fields back and fourth switches between

15:31 A and B won't notice these changes and will switch the assumed vc2s sale just

15:37 like the authentic vc2s sales in public packet switching Network U1 won't gain

15:44 too much by District however in an ATM network if quality of service is guaranteed then user one can gain a lot

15:53 of stealing a higher quality Channel which user one is not entitled to use ACC to the excess control policy user one

16:03 can gain even more if every user has to pay for the communication in both cases user two will be hurt someone may argue

16:12 that the possibility that the switches will compromises pretty low that will true if the ATM network is owned by one

16:20 organization when we consider ATM inter networking in which Kells will travel through different ATM networks it will

16:28 will be very easy for two switches to compromise traffic analysis traffic analysis refers to a threat that the

16:36 hacker can get information by collecting and analyzing the information like the volume timing and communication parties

16:44 of a VC volume and timing can reveal a lot of information to the hacker even though the data is encrypted because

16:52 encryption won't affect the volume and timing of information and also the source and this destination parties can be obtained from

17:00 the cell header normally is in clear text and some knowledge about the routing table another related thre is called

17:10 convert channels in this technique the attacker can encode the information in the timing and volume of data VCI or

17:19 even session key to release information to another people without being monitored normally these two attack

17:27 won't happen however when ATM is used in an environment requiring stringent security it might

17:35 happen traffic Management in order for ATM networks to deliver guaranteed quality of service on demand while

17:43 maximizing the utilization of available network resources effective traffic management mechanisms are needed almost

17:51 every aspect of ATM network operation from signaling requests and routing to network resource allocation and policing

17:59 contains some traffic management mechanisms some of parameters are as follows quality of service attributes

18:09 while setting up a connection on ATM networks users can specify the following parameters related to the desired

18:15 quality of service Peak cell rate PCR the maximum instantaneous rate at which

18:23 the user will transmit for bursty traffic the inter cell interval and the cell rate varies considerably the PCR is

18:32 the inverse of the minimum inter cell interval sustained cell rate

18:39 ISC this is the average rate as measured along a long time interval cell loss ratio CLR the

18:48 percentage of cells that are lost in the network because of error or congestion and are not delivered to the destination

18:55 recall that each ATM cell has a cell loss priority the CLP bit in the header

19:02 during periods of congestion the network will first discard cells with CLP is equal to 1 since the loss of cells with

19:11 CLP is equal to zero is more harmful to the operation of the application CLR can be specified separately for cells with

19:20 CLP is equal to 1 and for those with CLP is equal to zero cell transfer delay ctd

19:29 the delay experienced by a Cell between Network entry and exit points is called the sale transfer delay it includes

19:38 propagation delays queing delays at various intermediate switches and service times at queing

19:45 Point cell delay variation CDV this is mayor of variance of

19:52 ctd high variation implies larger buffering for delay sensitive traffic such as Vo and video burst tolerance

20:02 BT this determines the maximum burst size that can be sent at the peak rate this is the bucket size parameter for

20:10 leaky bucket algorithm that is used to control the traffic entering the network the algorithm consist of putting all

20:18 arriving cells in a buffer bucket which is drained at the sustained cell rate is

20:25 CR the maximum number of backto back cells that can be sent at the peak sell rate is called maximum burst size

20:34 MBS minimum sell rate MCR this is the minimum rate desired by a user the first six of the above

20:43 traffic parameters were originally specified in un version 3 traffic contract to provide a

20:52 guaranteed qos a traffic contract is established during connection setup which contains a connection traffic

21:00 descriptor and a confirmance definition however it is not necessary for every ATM virtual connection to have

21:08 a specified qos the reason for this is that if only specified qos connections are supported

21:16 by ATM then a large percentage of the network resources will be wasted this can happen when one or more connections

21:24 are not utilizing the full capacity of their qos cont contracts unspecified qos contracts can

21:32 be supported by an ATM network on a best effort basis such best effort services are sufficient for supporting most of

21:41 the existing data applications congestion control techniques congestion control lies at

21:49 the heart of the general problem of traffic management for ATM networks in general congestion arises when the

21:56 incoming traffic to a specified link is more than the outgoing link capacity the primary function of congestion control

22:04 is to ensure good throughput and delay performance while maintaining a fair allocation of network resources to the

22:11 user one method to avoid Network congestion is to accept a new ATM connection during connection setup phase

22:20 only when sufficient network resources are available to provide the acceptable qos this is called connection and

22:28 Mission Control CAC which is needed for connections where the qos must be

22:35 guaranteed disadvantages ATM has not been widely accepted although some phone companies

22:43 still use it in their backboard networks the expense complexity and lack of interoperability with other technologies

22:52 have prevented ATM from becoming more prevalent expense ATM technology provides a comprehensive list of

23:00 services even a moderate ATM switch costs much more than inexpensive land Hardware in addition the network

23:08 interface card needed to connect a computer to an ATM network is significantly more expensive than a

23:15 corresponding ethernet Nic connection setup latency ATM's connection oriented

23:23 Paradigm introduces significant delay for distant communication the time required to set up and tear down the ATM

23:30 VC for distant communication is significantly larger than the time required to use it

23:39 headers impose a 10% tax on all data transfer in case of ethernet sell taxes 1% lack of efficient

23:48 broadcast connection oriented networks like ATM are sometimes called non-broadcast multiple

23:56 exess NB M networks because the hardware does not support broadcast or multicast

24:05 on an ATM network broadcast to a set of computers is simulated by arranging for an application program to pass a copy of

24:14 data to each computer in the set as a result broadcast is inefficient complexity of

24:23 qos the complexity of this specification makes implementation cumbers some and difficult many implementations do not

24:32 support the full standard Assumption of homogeneity ATM is designed to be a

24:38 single Universal networking system there is minimal provision for interoperating

24:45 with other Technologies so friends this was all about ATM network security protocols hope the concepts explained in

24:53 this lecture were understandable and helpful hope to see you in the next lecture till then goodbye thank you

25:01 25 minutes, 1 second [Music]